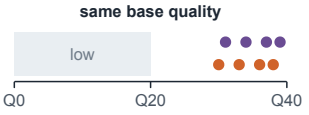
A Where the background comes from

Quality-based test (e.g. LoFreq, Geneious)

✓ correct ✕ wrong



mutation called — error miscalled

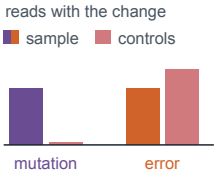


errors from PCR, homopolymers, paralog reads look like mutations

Control-calibrated test (SHMtrace)

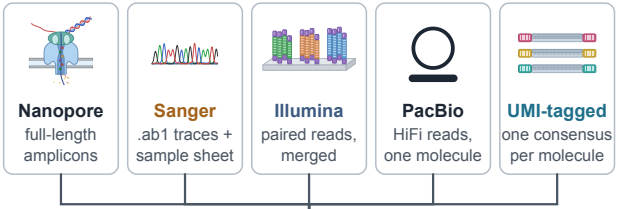


mutation called — error rejected

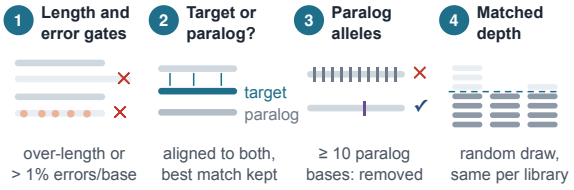


AID-null controls from the same run; empirical FDR target 0.05

B Inputs and read preparation



Read preparation



Sanger: trimming at Q ≥ 20 replaces 1; every clone is kept, so no 4

C Calling: a test for reads, a count for Sanger clones

Reads (nanopore, Illumina, UMI): tested against controls

1 Control error model



rate per type, strand, quality + held-out margin

2 Test and FDR



decoys (open) set the threshold: FDR target 0.05

Sanger clones: counted

1 Count the mismatches



every substitution at quality ≥ 20 is counted

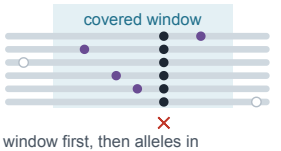
2 Background reported



shown beside the calls, never subtracted

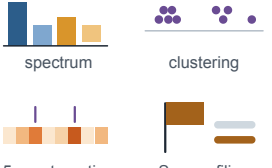
Both routes

3 Germline and window



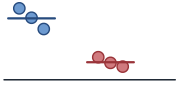
window first, then alleles in ≥ 50% of reads set aside and reported, not dropped, before any test; indels optional (opt-in: known ancestor)

4 Automatic checks



D Outputs

Animal-level tests and power



NB, Welch, permutation

Single-read estimate



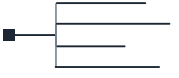
observed minus control background

Clones from the V(D)J junction



reads grouped by junction; each clone counted once

Mutation trees



full-length reads of one clone

Spectrum and motif shares



G:C / A:T, Ts / Tv, WRC/GYW, WA/TW

Reports, app and command line



HTML report, tables, project files